



Search ...



You are here: [Home](#) » [Switching](#)

Etherchannel on Cisco IOS Catalyst Switch ①

Lesson Contents

- 1. Configuration
 - 1.1. PAgP
 - 1.2. LACP
 - 1.3. Manual
 - 1.4. Load Balancing
- 2. Conclusion

In this lesson, we'll take a look at **EtherChannel**, which is also known as **link aggregation**. EtherChannel is a technology that lets you bundle multiple physical links into a single logical link. We'll take a look at how it works and what the advantages of EtherChannel are. Let's start with an example of a small network:

Course Contents

Switching

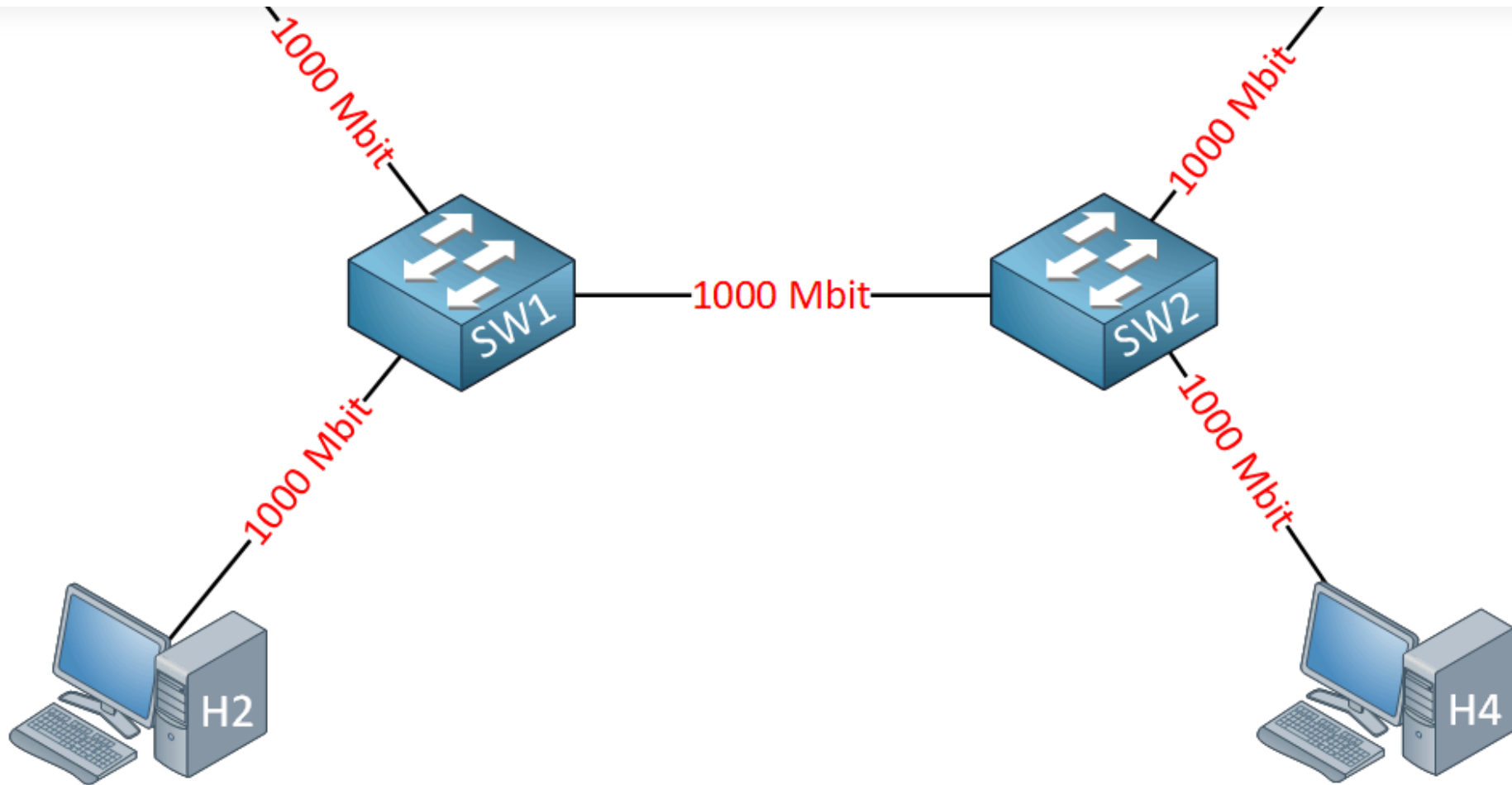
- + Unit 1: Switching Basics
- + Unit 2: VLANs and Trunking
- + Unit 3: VTP (VLAN Trunking Protocol)
- + Unit 4: Spanning-Tree
- + Unit 5: Etherchannel

Introduction to Etherchannel

Layer 3 Etherchannel

Troubleshooting Etherchannel

- + Unit 6: Virtualization
- + Unit 7: Design
- + Unit 8: Security



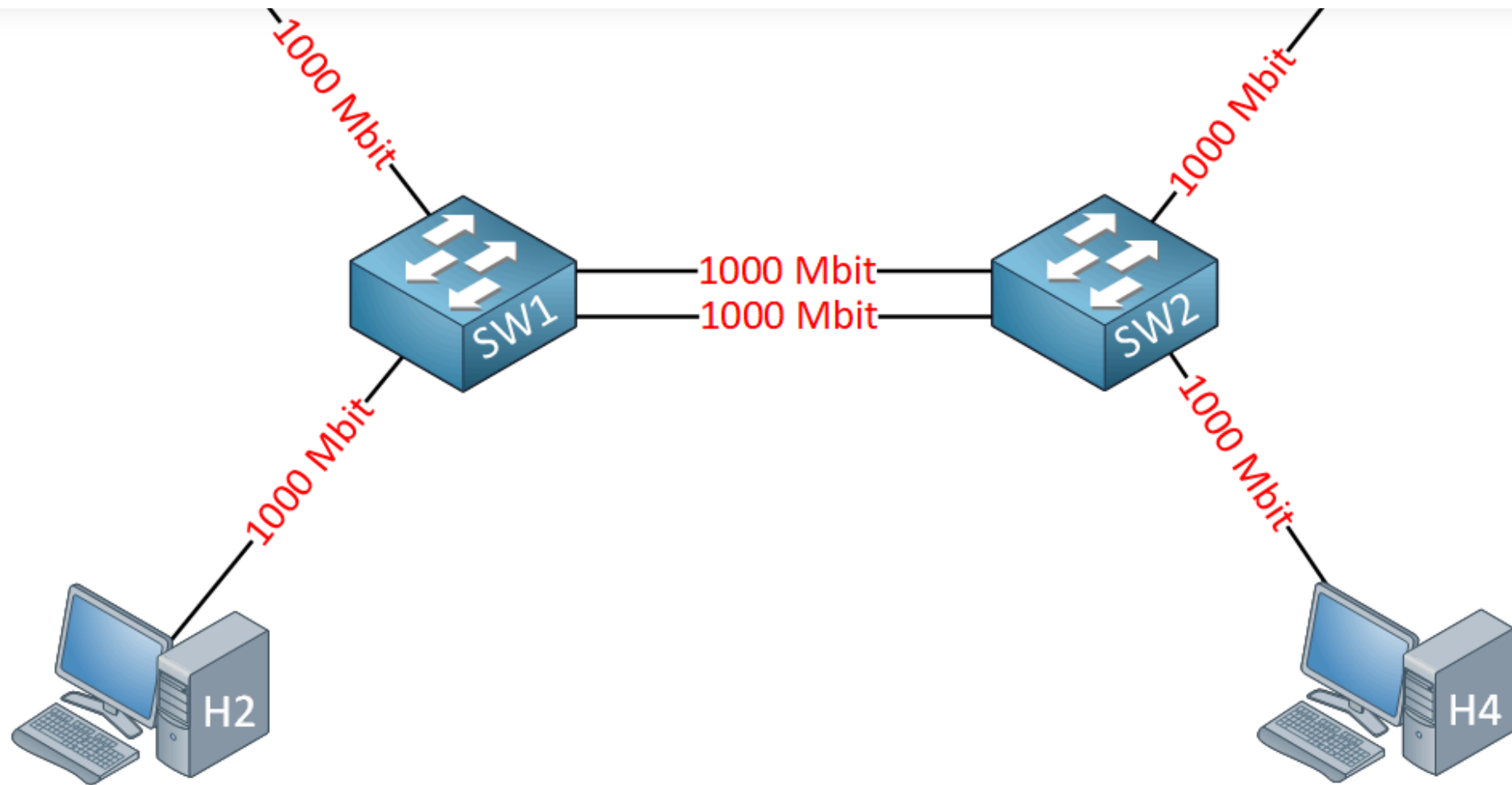
Take a look at the picture above. I have two switches and two computers connected to each switch. We use Gigabit Ethernet interfaces everywhere.

Now imagine that H1 sends 800 Mbit of traffic destined for H3 and H2 sends 600 Mbit of traffic destined for H4. The link between the switches will be a bottleneck. We require $800 + 600 = 1400$ Mbit, but we only have a 1000 Mbit link.

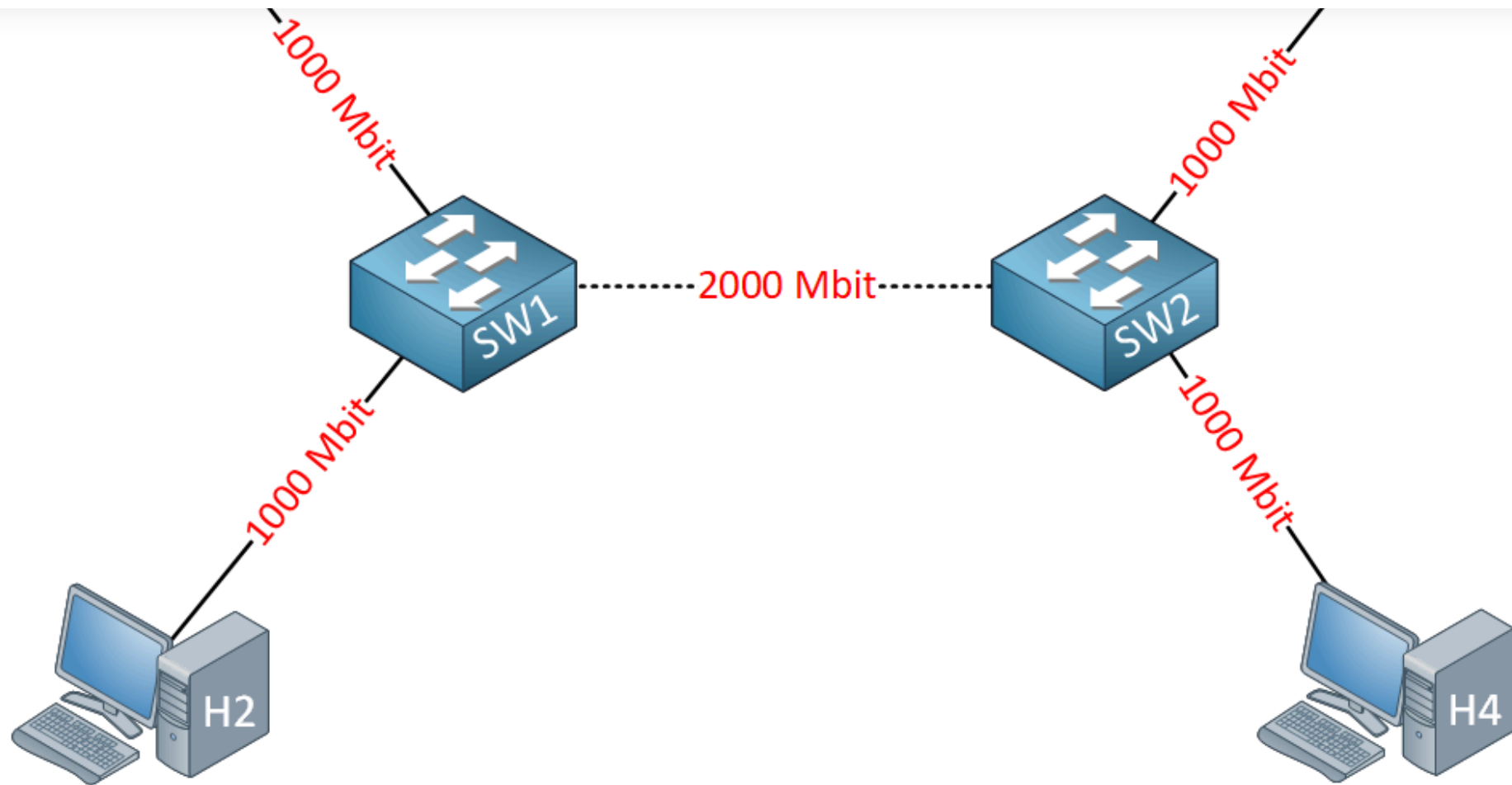


Since this lesson is about EtherChannel, we'll take a look at adding multiple links. Here's an example:

01:18



In the picture above, I have added a couple of extra links. The problem with this setup is that we have a loop between the switches, so spanning tree would block one out of two links. EtherChannel solves this problem because it creates a **single virtual link** out of these physical links:



Spanning tree sees this link as one logical link, so there are **no loops!** EtherChannel will do **load balancing** among the different links that we have, and it takes care of redundancy. Once one of the links fails, it will keep working and use the links that we have left.

Keep in mind that the physical links remain the limiting factor. A single traffic flow won't be able to exceed > 1000 Mbit (single Gigabit link). An Etherchannel is the equivalent of adding more lanes to a highway. The bandwidth increases, but



If you want to configure an EtherChannel then we have three options:

- **PAgP (Cisco proprietary)**
- **LACP (IEEE standard)**
- **Manual**

PAgP and LACP are negotiation protocols that dynamically configure an Etherchannel. PAgP is a Cisco proprietary protocol so you can only use it between Cisco devices. LACP is an IEEE standard which many vendors support.

It's also possible to configure a static EtherChannel without these protocols doing the negotiation of the link for you.

If you are going to create an EtherChannel you need to make sure that all interfaces have the same configuration:

- Duplex.
- Speed.
- Native and allowed VLANs.
- Switchport mode (access or trunk).

PAgP and LACP will check if the configuration of the interfaces that you use is the same.

1. Configuration

Let's see if we can configure an Etherchannel. We'll try all three options but start with PAgP.

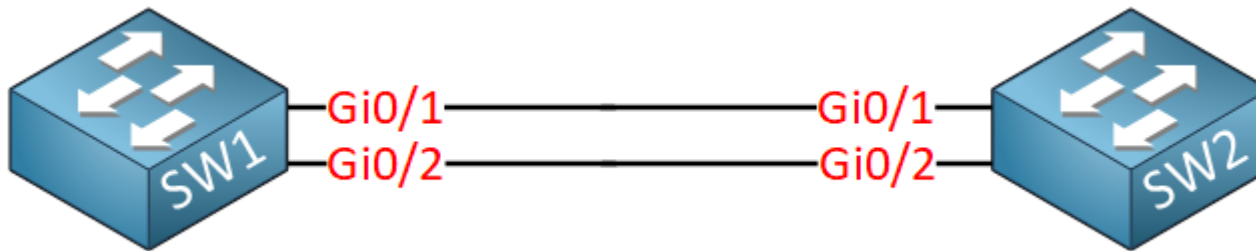
1.1. PAgP



Here's a quick overview with the two PAgP configuration options and whether an EtherChannel will form or not:

	Desirable	Auto
Desirable	Yes	Yes
Auto	Yes	No

Let me show you an example of how to configure PAgP between two switches. I'll use SW1 and SW2 for this demonstration:



SW1 and SW2 each have two Gigabit interfaces which we'll bundle these into a single logical link.

Here are our options:

```
SW1(config)#interface GigabitEthernet 0/1
SW1(config-if)#channel-group 1 mode ?
    active      Enable LACP unconditionally
    auto        Enable PAgP only if a PAgP device is detected
```



we use the `channel-group` command on the interface-level. You can pick any number you like, I'll go for 1. We need to decide whether we use auto or desirable mode. I'll configure SW1 to use desirable mode:

```
SW1(config)#interface range GigabitEthernet 0/1 - 2
SW1(config-if)#channel-group 1 mode desirable
```

I configure SW1 for PAgP desirable mode. It will actively ask SW2 to become an EtherChannel this way. We can configure SW2 in either auto or desirable mode. I'll use auto mode:

```
SW2(config)#interface range GigabitEthernet 0/1 - 2
SW2(config-if)#channel-group 1 mode auto
```

With auto mode, SW2 will only respond to incoming PAgP requests. After a few seconds, you'll see the following message on both switches:

```
SW1 %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
```

The switch automatically creates a new port-channel interface:

```
SW1#show interfaces port-channel 1
Port-channel1 is up, line protocol is up (connected)
  Hardware is EtherChannel, address is fa16.3e15.5fb2 (bia fa16.3e15.5fb2)
  MTU 1500 bytes, BW 2000000 Kbit/sec, DLY 10 usec,
```




```
Last clearing of "show interface" counters never
Input queue: 0/2000/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  12 packets input, 973 bytes, 0 no buffer
  Received 0 broadcasts (0 multicasts)
  0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  0 input packets with dribble condition detected
240 packets output, 17496 bytes, 0 underruns
  0 output errors, 0 collisions, 0 interface resets
  0 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
  0 output buffer failures, 0 output buffers swapped out
```

At the moment, there's nothing configured under the port-channel interface:

```
SW1#show run interface Port-channel 1
```

```
interface Port-channel1
end
```



```
interface GigabitEthernet0/1
  channel-group 1 mode desirable
end
```

```
SW1#show run interface GigabitEthernet 0/2
```

```
interface GigabitEthernet0/2
  channel-group 1 mode desirable
end
```

If you want to make any changes like configuring the EtherChannel as a trunk, you need to do this under the port-channel interface. For example:

```
SW1(config)#interface Port-channel 1
SW1(config-if)#switchport trunk encapsulation dot1q
SW1(config-if)#switchport mode trunk
```

When you do this, the switch automatically adds the commands to the physical interfaces as well:

```
SW1#show run interface GigabitEthernet 0/1
```

```
interface GigabitEthernet0/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
```



```
interface GigabitEthernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  channel-group 1 mode desirable
end
```

You shouldn't make any changes to the physical interfaces that belong to the EtherChannel. Always use the port-channel interface.

Let's continue and take a closer look at the status of the EtherChannel with some show commands. Here's the first one:

```
SW1#show etherchannel summary
```

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
```

```
        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators:          1
```



The `show etherchannel summary` command gives a nice clean overview with all your EtherChannels. It tells us whether the EtherChannel is up and which protocol we use. Here's another show command:

```
SW1#show etherchannel 1 port-channel
```

```
Port-channels in the group:
```

```
-----
```

```
Port-channel: Po1
```

```
-----
```

```
Age of the Port-channel    = 0d:00h:10m:16s
```

```
Logical slot/port    = 2/1          Number of ports = 2
```

```
GC                    = 0x00010001    HotStandBy port = null
```

```
Port state            = Port-channel Ag-Inuse
```

```
Protocol              = PAgP
```

```
Port security         = Disabled
```

```
Ports in the Port-channel:
```

Index	Load	Port	EC state	No of bits
0	00	gi0/1	Desirable-S1	0
0	00	gi0/2	Desirable-S1	0



it tells us whether the logical interface is up.

Here's the last show command you can use:

```
SW1#show interfaces GigabitEthernet 0/1 etherchannel
```

```
Port state      = Up Mstr In-Bndl
```

```
Channel group = 1          Mode = Desirable-S1    Gcchange = 0
```

```
Port-channel  = Po1        GC    = 0x00010001    Pseudo port-channel = Po1
```

```
Port index     = 0          Load = 0x00          Protocol = PAgP
```

```
Flags: S - Device is sending Slow hello.  C - Device is in Consistent state.
```

```
      A - Device is in Auto mode.          P - Device learns on physical port.
```

```
      d - PAgP is down.
```

```
Timers: H - Hello timer is running.        Q - Quit timer is running.
```

```
      S - Switching timer is running.      I - Interface timer is running.
```

Local information:

Port	Flags	State	Timers	Hello Interval	Partner Count	PAgP Priority	Learning Method	Group Ifindex
Gi0/1	SC	U6/S7	H	30s	1	128	Any	5001

Partner's information:

Port	Partner Name	Partner Device ID	Partner Port	Partner Age	Partner Flags	Partner Group Cap.
------	--------------	-------------------	--------------	-------------	---------------	--------------------



The third method to verify your EtherChannel is to use the `show interfaces` command. In my example, I am looking at the information of the GigabitEthernet 0/1 interface. Besides information on our local switch, you can also see the interface of our neighbor switch.

Configurations

Want to take a look for yourself? Here you will find the final PAgP configuration of each device.

SW1

```
hostname SW1
!
interface GigabitEthernet0/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  channel-group 1 mode desirable
!
interface GigabitEthernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  channel-group 1 mode desirable
!
interface port-channel 1
  switchport trunk encapsulation dot1q
  switchport mode trunk
```



```
hostname SW2
!
interface GigabitEthernet0/1
  channel-group 1 mode auto
!
interface GigabitEthernet0/2
  channel-group 1 mode auto
!
interface port-channel 1
!
end
```

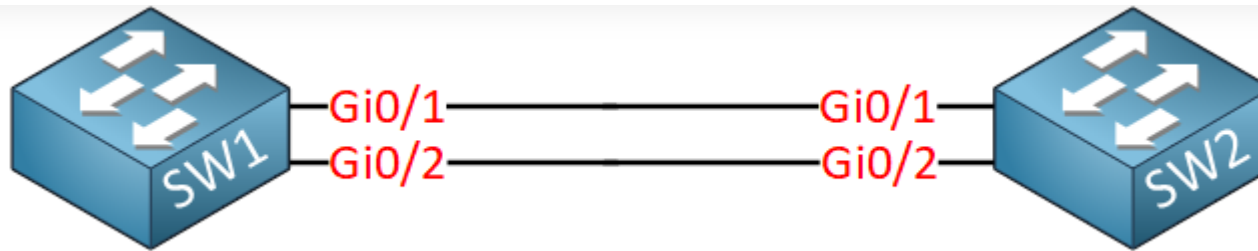
1.2. LACP

LACP is similar to PAgP but uses different terminology:

- **Active:** The interface will actively ask the other side to become an EtherChannel.
- **Passive:** The interface waits passively for the other side to ask to become an EtherChannel.

At least one switch should use active mode. When both switches use passive mode, nothing will happen.

	Active	Passive
Active	Yes	Yes



We'll configure SW1 to use active mode:

```
SW1(config-if)#interface range GigabitEthernet 0/1 - 2
SW1(config-if)#channel-group 1 mode active
```

And SW2 will use passive mode:

```
SW2(config)#interface range GigabitEthernet 0/1 - 2
SW2(config-if)#channel-group 1 mode passive
```

Let's verify our work. We can use the same show commands we use for PAgP:

```
SW1#show etherchannel summary

Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
```




w - waiting to be aggregated

d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gi0/1(P) Gi0/2(P)

Let's try the show etherchannel command:

SW1#show etherchannel 1 port-channel

Port-channels in the group:

Port-channel: Po1 (Primary Aggregator)

Age of the Port-channel = 0d:00h:01m:43s



Load share deferral = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
0	00	Gi0/1	Active	0
0	00	Gi0/2	Active	0

Time since last port bundled: 0d:00h:01m:04s Gi0/2

And last but not least, let's try show interfaces:

SW1#show interfaces GigabitEthernet 0/1 etherchannel

Port state = Up Mstr Assoc In-Bndl

Channel group = 1 Mode = Active Gcchange = -

Port-channel = Po1 GC = - Pseudo port-channel = Po1

Port index = 0 Load = 0x00 Protocol = LACP

Flags: S - Device is sending Slow LACPDUs F - Device is sending fast LACPDUs.

A - Device is in active mode. P - Device is in passive mode.

Local information:

LACP port	Admin	Oper	Port	Port
-----------	-------	------	------	------



LACP port					Admin	Oper	Port	Port
Port	Flags	Priority	Dev ID	Age	key	Key	Number	State
Gi0/1	SP	32768	5e00.0001.8000	18s	0x0	0x1	0x2	0x3C

Age of the port in the current state: 0d:00h:01m:37s

As you can see the protocol is now LACP, we have a Port-channel 1 interface with two member physical interfaces.

Configurations

Want to take a look for yourself? Here you will find the final LACP configuration of each device.

SW1

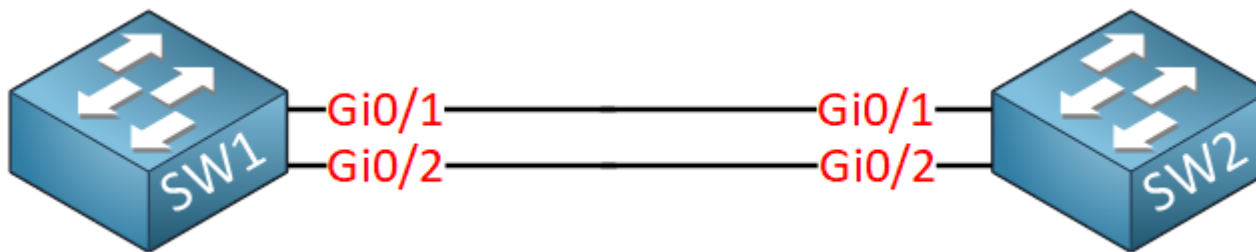
```
hostname SW1
!
interface GigabitEthernet0/1
  channel-group 1 mode active
!
interface GigabitEthernet0/2
  channel-group 1 mode active
!
interface port-channel 1
```



```
hostname SW2
!
interface GigabitEthernet0/1
  channel-group 1 mode passive
!
interface GigabitEthernet0/2
  channel-group 1 mode passive
!
interface port-channel 1
!
end
```

1.3. Manual

Instead of PAgP or LACP, we can also manually enable the Etherchannel. I'll use the same topology to demonstrate this:



Here's the configuration:



```
SW2(config-if-range)#channel-group 1 mode on
```

That's all there is to it. Let's try our show commands. We'll start with an overview:

```
SW1#show etherchannel summary
```

```
Flags:  D - down          P - bundled in port-channel
```

```
        I - stand-alone s - suspended
```

```
        H - Hot-standby (LACP only)
```

```
        R - Layer3        S - Layer2
```

```
        U - in use        N - not in use, no aggregation
```

```
        f - failed to allocate aggregator
```

```
        M - not in use, minimum links not met
```

```
        m - not in use, port not aggregated due to minimum links not met
```

```
        u - unsuitable for bundling
```

```
        w - waiting to be aggregated
```

```
        d - default port
```

```
        A - formed by Auto LAG
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```



In the output above, you can see we don't use any protocol. Let's take a closer look at the Port-channel interface.

```
SW1#show etherchannel 1 port-channel
```

```
Port-channels in the group:
```

```
-----
```

```
Port-channel: Po1
```

```
-----
```

```
Age of the Port-channel    = 0d:00h:06m:54s
```

```
Logical slot/port    = 16/0          Number of ports = 2
```

```
GC                    = 0x00000000    HotStandBy port = null
```

```
Port state            = Port-channel Ag-Inuse
```

```
Protocol              = -
```

```
Port security         = Disabled
```

```
Load share deferral = Disabled
```

```
Ports in the Port-channel:
```

Index	Load	Port	EC state	No of bits
0	00	Gi0/1	On	0
0	00	Gi0/2	On	0



```
SW1#show interfaces GigabitEthernet 0/1 etherchannel
```

```
Port state      = Up Mstr In-Bndl
```

```
Channel group = 1          Mode = On          Gcchange = -
```

```
Port-channel   = Po1      GC    =    -      Pseudo port-channel = Po1
```

```
Port index     = 0        Load = 0x00      Protocol =    -
```

```
Age of the port in the current state: 0d:00h:01m:41s
```

That's all we have.

Configurations

Want to take a look for yourself? Here you will find the final manual configuration of each device.

SW1

```
hostname SW1
```

```
!
```

```
interface GigabitEthernet0/1
```

```
  channel-group 1 mode on
```

```
!
```

```
interface GigabitEthernet0/2
```

```
  channel-group 1 mode on
```



SW2

```
hostname SW2
!
interface GigabitEthernet0/1
  channel-group 1 mode on
!
interface GigabitEthernet0/2
  channel-group 1 mode on
!
interface port-channel 1
!
end
```

1.4. Load Balancing

The last thing I want to show you about EtherChannel is load-balancing. Take a look at the following output:

```
SW1#show EtherChannel load-balance
EtherChannel Load-Balancing Configuration:
      src-mac

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source MAC address
```




balances based on the source MAC address. If you want, you can change this behavior with the `port-channel load-balance` command:

```
SW1(config)#port-channel load-balance ?
```

```
dst-ip      Dst IP Addr
```

```
dst-mac     Dst Mac Addr
```

```
src-dst-ip  Src XOR Dst IP Addr
```

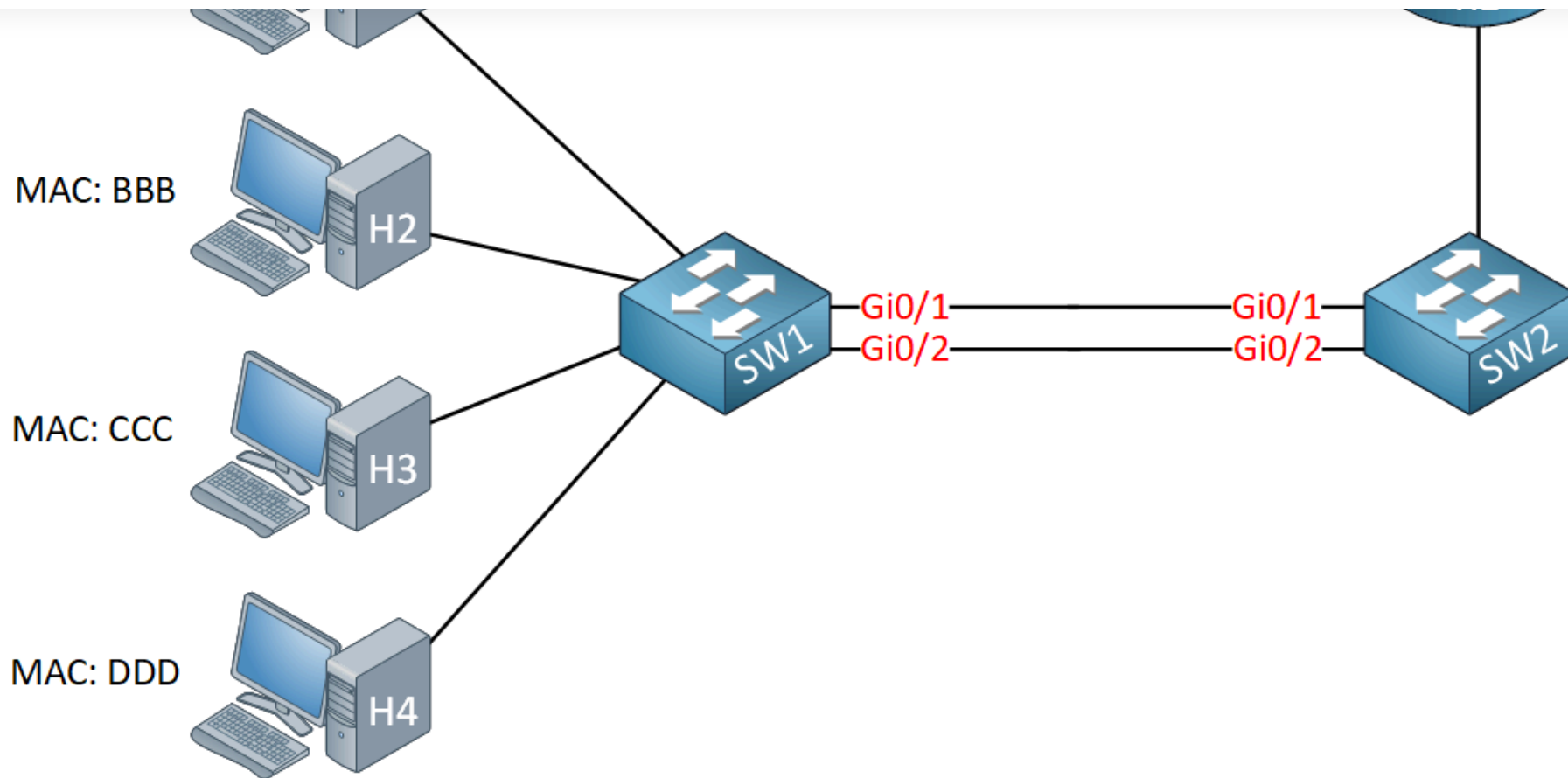
```
src-dst-mac Src XOR Dst Mac Addr
```

```
src-ip      Src IP Addr
```

```
src-mac     Src Mac Addr
```

There are plenty of options to choose from, including combinations of source and/or destination MAC or IP addresses.

Why should you care about load balancing? Take a look at the picture below:



We have SW1 and four computers. On the right side, we have SW2 and a router. The default load-balancing mechanism is the *source MAC address*. This means that **ALL traffic from one MAC address** will be sent down one and the same physical interface, for example:

- MAC address AAA will be sent using SW1's gi0/1 interface.
- MAC address BBB will be sent using SW1's gi0/2 interface.
- MAC address CCC will be sent using SW1's gi0/1 interface.



It's a different story for SW2 since we only have one router with MAC address EEE. It will pick one of the physical interfaces so ALL traffic from the router will be sent down interface GigabitEthernet 0/1 or 0/2. One of the physical links won't be used at all.

If this is the case, it might be better to change your load-balancing algorithm. For example, on SW2, we could do this:

```
SW2(config)#port-channel load-balance dst-mac
```

If we switch the load balancing to destination MAC address on SW2 then traffic from our router to the computers will be load-balanced amongst the different physical interfaces because we have multiple computers with different destination MAC addresses.

Configurations

Want to take a look for yourself? Here you will find the final load-balancing configuration of each device.

SW1

```
hostname SW1
!
interface GigabitEthernet0/1
  channel-group 1 mode active
!
interface GigabitEthernet0/2
  channel-group 1 mode active
!
interface port-channel 1
```



SW2

```
hostname SW2
!
interface GigabitEthernet0/1
  channel-group 1 mode passive
!
interface GigabitEthernet0/2
  channel-group 1 mode passive
!
interface port-channel 1
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
port-channel load-balance dst-mac
!
end
```

That's all I have on EtherChannels for now; hopefully, this has been helpful to you! If you have any questions feel free to leave a comment.

2. Conclusion

You have now learned what EtherChannels are and how to configure them:



- Manual: Bundle multiple physical interfaces without any protocol.
- EtherChannels support multiple load-balancing options. Make sure you select a load-balancing algorithm that matches your traffic patterns.

I hope you enjoyed this lesson. If you have any questions, please leave a comment.

◀ **Previous Lesson**
FlexLinks

Next Lesson ▶
Layer 3 Etherchannel

🔖 Tags: Etherchannel, IOS, LACP, Link Aggregation, PAGP

Forum Replies



ReneMolenaar

Hi, Rene. Thanks for info. What if you add note about pagp silent/non-silent submode? Thanks.



ReneMolenaar



Rene



281332

hi Rene, i can't understand what's different between PAgP and LACP, why cisco created PAgP? does it has some pros?



ReneMolenaar

Hi Yevgeniy,

Good question. Etherchannel was originally invented by a company called Kalpana in the 90s. Cisco bought the company in 1994 and released PaGP. Later in 2000, IEEE released an open standard version for Etherchannel (802.3ad) which is LACP.

Rene



hussien.samer

Hi Rene,

What is the benefit of etherchannel protocols if we can configure interface as etherchannel manually by setting the mode of etherchannel (on) in both side ? so why we need to set one side Desirable or Active and the other side Auto or Passive ?

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